



Grounds for improving the implementation of game-oriented forest management – A double sampling survey of Finnish forest owners and professionals

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ABSTRACT

The aim of the study was to evaluate the current level of awareness, implementation and need for knowledge of game-oriented forest management (GOFM) among forest owners and professionals. This information is necessary to understand the grounds for improving awareness and implementation of new forest management guidelines and to evaluate suitable policy instruments for the future. The study joins the body of research on innovation diffusion and institutional adaptation, in regard to new forest management methods. Our results from Finland indicated that a high proportion of forest owners have not adopted GOFM, despite the information provided. A wider implementation of the method might require other knowledge dissemination methods, such as providing tangible examples and peer learning to forest owners. The role of forest professionals as change agents was seen to be important for the generalization of GOFM among forest owners and inside institutions. The observed level of awareness of GOFM and its implementation among forest owners, and the engagement of professionals reflects a situation where the informing process is ongoing, and where the forestry advisors need tools to engage forest owners more effectively. At the same time, it is difficult to estimate the effectiveness of the voluntary-based approach of GOFM. In this study, the double sampling design removed volunteer bias and increased the reliability of the results.

1. Introduction

1.1. Justification of game-oriented forest management

In forest management, a variety of strategies and methods have been suggested and developed for managing wildlife and habitats (e.g. Askins, 2001; Dayer et al., 2014; Rautiainen et al., 2017; Dressel et al., 2018). In Finland, game-oriented forest management (GOFM) was created to promote biodiversity, and the wellbeing of game birds and animals in commercially used forests. The grouse species that live in the boreal forests of Fennoscandia (capercaillie *Tetrao urogallus*, black grouse *Tetrao tetrix*, hazel grouse *Tetrastes bonasia*, willow grouse *Lagopus lagopus*) are especially valued as game species, but their long-term population trends have been rather negative, due to increased predator

densities, hunting, and climatic changes, and especially due to the detrimental effects of intensive forestry on their forest and mire habitats (e.g. Rolstad and Wegge, 1989; Kurki et al., 2000; Miettinen et al., 2008; Rautiainen et al., 2017; Melin et al., 2020). As such, the specific emphasis in GOFM has been on the habitat management of grouse, as moose (the other important forest game species) tend to manage well in the younger forests typical of the landscapes of Fennoscandia (e.g. Edenius et al., 2002; Lindén et al., 2014; Dressel et al., 2018).

Although habitat requirements vary between grouse species, rich field-layer vegetation and dwarf shrubs, especially bilberry (*Vaccinium myrtillus*), are characteristic of grouse habitats (e.g. Melin et al., 2016; Rautiainen et al., 2017). Mixed tree composition is also important, and spruce mire environments and forest-mire ecotones are key habitats (e.g. Lindén et al., 2014; Rautiainen et al., 2017). In GOFM, the

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emphasis is on acknowledging these characteristics and habitats throughout all forest operations (stand treatments, thinnings, cuttings etc.), whereas the common forest management methods currently employed in Fennoscandia have mostly aimed at growing even-aged forest stands with a single tree species for wood production. This has simplified forest structures and has affected the shrub-layer vegetation (e.g. Östlund et al., 1997; Vanha-Majamaa, 2001; Edenius et al., 2002). Regardless of the strong emphasis on grouse in GOFM, habitats suitable for grouse are often valuable for other game species (e.g. hare) and for biodiversity generally. For example, the grouse species capercaillie is regarded as an umbrella species in Europe, and its status can also reflect the wellbeing of other species (e.g. pygmy owl, three-toed woodpecker, red breasted flycatcher) (Suter et al., 2002; Pakkala et al., 2003; Mikoláš et al., 2015).

For forest owners, the benefits of utilizing GOFM is related to its multifunctional nature and cost-efficiency. Applying GOFM would be valuable for the forest owner, who could maintain game habitats in her/his forests, but still utilize the forest for other purposes, such as wood production. GOFM does not restrict the management of forests aside from preserving shelter and food for game species during forest operations. Indeed, in most cases the measures are straightforward and easy to implement (e.g. maintaining game thickets and undergrowth).

1.2. Game-oriented forest management as a soft policy instrument

In Finland, habitat-oriented forest management practices have been available for decades; forest management methods with similar objectives were introduced in the 1980s (Metsähallitus, 1989; Rautiainen et al., 2017). However, due to a strongly production-oriented forest sector, these methods were never fully taken up by non-industrial private forest owners (hereafter referred to as forest owners) (Rautiainen et al., 2017), who own more than 60% of Finnish forest land (e.g. Hänninen et al., 2011; Natural Resources Institute Finland, 2019). More than 80% of the roundwood purchased annually by the Finnish forest industry is produced on private forest land (Hänninen et al., 2011; Parviainen and Västilä, 2011), which might have contributed to the production-oriented forest management behavior of forest owners. At the same time, forest owners have also relied considerably on the assistance and expertise of forest professionals in forest-related decision-making (e.g. Hujala and Tikkanen, 2008; Hannerz et al., 2010; Hänninen et al., 2011; Valkonen and Cheng, 2014), which also highlights the role of professionals and organizations in forest management on private land. Forest workers and organizations have been central in enhancing the adoption of forestry approaches (Muth and Hendee, 1980), and a generalization of a method requires its assimilation, not only among forest owners, but also among the professionals and institutions that assist the forest owners in forest planning.

Currently, the majority of forest owners consider themselves multi-objective (Hallikainen et al., 2010; Hänninen et al., 2011), and considerable changes in Finnish forest legislation (2014) have now provided forest owners with better opportunities to manage their forests with alternative management methods (e.g. Ojala and Mäkelä, 2013). For example, GOFM, which can be considered as a forest management innovation, has been promoted more actively in recent years with published habitat and forest management guidelines and working instructions (e.g. Päivinen et al., 2011; Lindén et al., 2014; Finnish Wildlife Agency, 2015), especially by the Finnish Wildlife Agency and the Finnish Government owned Forest Management Advisory and Consulting Service Provider Tapio (Tapio, 2019). In addition, the Grouse Management Plan published in 2014 (Ministry of Agriculture and Forestry, 2014) presented forest management recommendations that were similar to GOFM.

The approaches outlined above may be viewed as “sermon” type forest policy instruments for the promotion of GOFM, but are also associated with financial instruments and market-based mechanisms. As a recent improvement, the preservation of game thickets will not

disqualify forests from subsidies under the Act on the Financing of Sustainable Forestry (this had previously led to disqualification). Game thickets have also been acknowledged by the forest certification systems, although certification does not directly contribute the implementation of GOFM. A clearer reference to GOFM in the certification criteria could create a nuance of game-orientation in forest management, promote habitat management and support biodiversity in production forests.

One of the strengths of the GOFM method, and in its generalization, could be its voluntary nature, and that all game-oriented operations in forests are based on the willingness of forest owners to participate. The voluntary-based working principle has been inspired by the positive experiences from the attitudes of forest owners in the voluntary forest biodiversity program METSO, following command-and-control types of policy mechanisms (Paloniemi and Tikka, 2008; Salomaa et al., 2016). However, research related to the implementation of GOFM has hitherto been very narrow and GOFM has been considered as a unique management method in Finnish forests. There is the assumption that, despite the provision of information, forest stakeholders have not yet adopted GOFM, and that more could be done in forest management to enhance the living conditions of Finnish game species by better utilizing new possibilities to implement GOFM. Previous studies that have evaluated the adoption of forestry approaches by Finnish forest owners (e.g. Korhonen et al., 2013) have introduced other knowledge dissemination methods for efficient adoption, such as peer learning and persuasion with “change agents” (see Rogers, 1995), who help forest owners to adapt their behavior to utilize new approaches.

1.3. Research objectives

The aim of this study is to evaluate (through questionnaire surveys to forest owners and professionals) the current state of GOFM; especially, (1) the extent of adoption and implementation by forest stakeholders, and (2) to what extent have the currently applied forest policy instruments (providing information) for GOFM implementation been successful. More specifically, we evaluate the current level of awareness, implementation and need for knowledge of GOFM among forest owners and professionals, which also represents the present integration of GOFM in forest management. This baseline information is important in planning future education and communication regarding GOFM. It is utilized to evaluate whether continuing to provide general GOFM information would result in an increase in the number of owners implementing GOFM, or whether other knowledge dissemination methods, such as persuasion and peer learning (see Korhonen et al., 2013), be introduced in order to promote GOFM implementation. We approach forest management from a game perspective, because Finnish forest owners tend to appreciate game more than biodiversity as a justification to modify forest management practices or to protect part of their forests (e.g. Hallikainen et al., 2010; Hujala et al., 2010).

2. Conceptual orientation

We approach the research objectives from two analytical aspects: innovation adoption and institutional adaptation. The former is inspired by Roger's (1995) innovation diffusion theory, in particular the role of advising change agents, peer support and the provision of tangible examples of the behavior of late adopters; see Korhonen et al. (2013) for adoption of voluntary biodiversity conservation among forest owners. Rogers (1995) described five adopter categories (innovators, early adopters, early majority, late majority, laggards), which are based on the time of innovation adoption and the decision process that underpins it. According to Rogers (1995), innovators account for 2.5%, and early adopters 13.5% of adopters, whereas early and late majority both account for 34%, and laggards 16% of adopters. The time used for innovation-decision and the need for “persuasion” and peer support for adoption increase toward the latter categories (as we would

also expect with GOFM, an innovation waiting to be adopted by forest owners). Usually, after the initial efforts, only the minority adopts the innovation and the diffusion intensifies once the adopters have been imitated by others (Muth and Hendee, 1980; Korhonen et al., 2013). However, adoption can be accelerated by change agents (professionals or forest workers), who affect the decision-making process (Muth and Hendee, 1980; Rogers, 1995; Korhonen et al., 2013).

According to Moore (2006), innovators and early adopters have substantially different expectations in regard to innovations than the latter adopter groups. Therefore, in our conceptual framing, we follow the idea proposed by Korhonen et al. (2013), and assume that early and late majority adopters, which are the largest categories according to Rogers (1995), shape the group of late adopters. We utilize this concept of different innovation adoption mechanisms by early and late adopters in our study, and assess their different requirements for encouragement, support or persuasion in innovation adoption in order to evaluate the state of GOFM implementation, the suitability of current forest policy instruments, and future policy implications.

Institutional adaptation borrows from the analyses of forest biodiversity conservation carried out by Primmer and Wolf (2009) and Primmer (2011). Based on those studies, our work presumes a uniformity of professional norms among forest professionals, which creates inertia in adopting new forest management methods, as well as the important role of strategically oriented organizations, which enable adaption to policy and management innovations. Conceptually, this study evaluates the integration of ecologically justified forest management recommendations for practical use in Finland. We observe the results of GOFM adoption by forest owners and professionals through an analytical lens, which contains information, advice, communication, and service delivery implications based on the observed levels of awareness, adoption, and interests. Thus, our conceptual framework, which encompasses the different types of adopters, change agents and institutions' adaptation to GOFM, directs our discussion of the relevance of current policy instruments and the future prospects for GOFM promotion.

3. Materials and methods

3.1. Sample and data collection protocol

In this study, we employed a double sampling method, which was first introduced by Neyman (1938) and is now found in most textbooks that describe standard sampling techniques (e.g. Cochran, 1977; Särndal et al., 1992; Thompson, 2012). We applied a variation of this method, where a sample of non-respondents was taken (as a second sample) and contacted again after the survey (e.g. Särndal et al., 1992; Fowler Jr. et al., 2002; Jenkins et al., 2004; Thompson, 2012) with a more effective method than in the original sample. The results from the second-phase sample were combined with the initial survey (first-phase sample) results to provide unbiased estimates of the population means.

The study was conducted through surveys targeted at Finnish forest owners and forest professionals. Contact information was requested as simple random samples from the contact information registers of the Finnish Forest Centre, which is part of an indirect state administration that works under the strategic guidance of the Ministry of Agriculture and Forestry of Finland. The Finnish Forest Centre maintains a register of private forest owners and holds another register of forest professionals, which includes employees from forest industry enterprises and associated forest services, forest governance, forestry societies, private forest services, contractors, forest education and officials. The sampling frames were databases of those non-industrial private forest owners and forest professionals who had provided email addresses as contact information, but no other delimiters were set to direct the sampling. A total of 2600 contacts were provided (1300 persons per group).

The data were collected in spring and summer 2017. Survey questions were designed to measure the substantive frequency of issues that

were (a) considered to reflect the current state of GOFM implementation and the barriers to implementation, and (b) needed to create strategies for the support of the forest stakeholders that might implement GOFM. These assumptions related to the adoption of GOFM were created by thematic interviews with forest professionals who have been working with GOFM, and who were able, therefore, to provide general views of the state of GOFM. In these interviews, forest stakeholder awareness in regard to GOFM and GOFM-related knowledge levels was questioned. A lack of awareness and, possibly, poor knowledge of GOFM were seen as affecting GOFM implementation, which was also influenced by the interactions between forest owners and professionals in GOFM-related decision-making, and were also connected to the role of professionals as change agents. Therefore, these themes were used as the framework for the survey forms. In the surveys, a definition of GOFM was not offered to avoid providing the respondents with pre-conceptions of the topic.

Survey forms were first tested with smaller samples (20 persons) from both groups with a protocol identical to that used in the initial surveys, in order to detect and modify possible complications in the questions or their presentation. The initial surveys were sent to two respondent groups (forest owners, forest professionals) of 1280 persons using an online system called SurveyPal. The respondents were given approximately three weeks response time and a reminder was sent each week during that time. After the surveys were closed, the data was transferred from the survey program to Microsoft Office Excel.

A total of 33.2% of forest owners ($n = 425$) and 37.5% of forest professionals ($n = 480$) participated in the survey (see background information in Table 1). These groups are hereafter referred to as the respondents, and those that did not take part in the survey are referred to as non-respondents. The response rates were similar to the average response rates for Finnish forest surveys (e.g. Valkeapää et al., 2009; Asikainen et al., 2014; Mattila et al., 2015; Toppinen et al., 2015).

After the first survey, the second phase sample was collected to obtain information about the attitudes of the non-respondents, and to minimize attitude-related volunteer bias from the results. The sample size was rather small, because resources for it were limited, and the data collection method through telephone calls was more time-consuming. The questions were asked from randomly selected non-respondents; 15 forest owners and 16 forest professionals (hereafter referred to as callback respondents). The questions were posed to whoever answered the phone; no one refused to answer the questions once they were informed of the reason for the call. We assume that the decision to answer a call from an unknown number was not associated with the person's attitude to GOFM, so we therefore consider that no volunteer bias was present in the second survey data. Therefore, the information from the second phase sample can be generalized to the whole population of non-respondents without bias.

Table 1

Average age and gender distribution of forest owners and professionals in this study. Forest owners' information is based on the contact register provided by Finnish Forest Center, and professionals' information is based on the responses of professionals in the survey. There were 1 (0.2%) missing age and gender entry among forest owners, and 9 (1.9%) missing age entries and 25 (5.2%) missing gender entries among professionals.

	Survey samples		Respondents	
	Forest owners ($N = 1280$)	Forest professionals ($N = 1280$)	Forest owners ($n = 425$)	Forest professionals ($n = 480$)
Average age	57 years	–	60 years	49 years
Female	317 (24.8%)	–	95 (22.4%)	55 (11.5%)
Male	962 (75.2%)	–	330 (77.6%)	400 (83.3%)

Table 2

Results from the game-oriented forest management (GOFM) questions posed to forest owners. $\bar{y}_d$ is the unbiased estimate of the population mean (combined frequency), and s_d its standard error. CI, LB and CI, UB represent the lower and upper bounds of confidence interval (95%) of the population mean $\bar{y}_d$. On the right-hand side, $\bar{y}_1$ (respondent population mean) represents the proportion of responses in the initial survey and $\bar{y}_2$ (non-respondent population mean) the proportion of responses in the callback survey.

		$\bar{y}_d$, %	s_d , %	CI, LB	CI, UB	$\bar{y}_1$, %	$\bar{y}_2$, %
Are you interested in implementing GOFM in your forests?	Yes	41.9	8.7	25.0	58.9	60.6	33.3
	No	58.1	8.7	41.2	75.1	39.4	66.7
Are you implementing or have you implemented GOFM in your forests?	Yes	40.0	8.9	22.6	57.4	40.0	40.0
	No	60.0	8.9	42.6	77.4	60.0	60.0
Are you aware of GOFM?	Yes	60.2	8.8	42.9	77.5	60.7	60.0
	No	39.8	8.8	22.5	57.1	39.3	40.0
Is GOFM information available?	Yes	58.3	8.5	41.6	75.0	41.1	66.7
	No	4.7	0.6	3.5	5.9	14.4	0
	I don't know	37.0	8.5	20.3	53.7	44.5	33.3
Would you like to have more GOFM information?	Yes	39.7	8.5	23.0	56.5	53.0	33.3
	No	60.3	8.5	43.5	77.0	47.0	66.7
Does your basic GOFM knowledge level affect your motivation to implement GOFM?	Yes	35.1	8.1	19.3	51.0	53.2	26.7
	No	64.9	8.1	49.0	80.7	46.8	73.3
Are GOFM services, such as planning, consulting, implementation etc. available?	Yes	21.9	7.3	7.6	36.2	25.9	20.0
	No	18.3	7.3	4.0	32.6	14.7	20.0
	I don't know	59.8	8.9	42.3	77.3	59.4	60.0
Have you received education or guidance on GOFM?	Yes	5.3	0.6	4.0	6.5	16.0	0
	No	94.8	0.6	93.5	96.0	84.0	100.0

3.2. Analysis

The unbiased estimates (or combined frequency rates) were computed by combining the answers of the initial respondents and callback respondents, as volunteer bias was considered to be absent in the second survey data. If only the original respondents were used for analysis without correction for volunteer bias, the estimates would have been very different and the confidence intervals much narrower. Narrow confidence intervals are more desirable than wide ones, but in this case, we acknowledged that the naïve estimates were simply wrong, and that narrower confidence intervals would have required a larger callback sample size. The volunteer bias could be reduced by grouping the respondents according to age, gender and other background variables, and thereafter weighting the results by the so-called propensity scores in each class (Thompson, 2012). However, even this approach assumes that the willingness to answer is similar among the non-respondents of each class than in the class of respondents, and there would still be an unknown amount of volunteer bias in the results.

The unbiased estimates of the population means and their variances, standard errors and confidence intervals were calculated. The unbiased estimates of population means $\bar{y}_d$ were provided by double sampling stratified estimator (Thompson, 2012):

$$\bar{y}_d = w_1 \bar{y}_1 + w_2 \bar{y}_2$$

where $w_1 = n_1'/n'$ is the weighting of the respondents and $w_2 = n_2'/n'$ is the weighting of non-responders; n_1' and n_2' are the number of respondents and non-respondents in the initial survey; n' is the sample selected from the total population (1280); $\bar{y}_1$ is the estimate of the respondent population mean (from the initial survey) and $\bar{y}_2$ the estimate of the non-respondent population mean from the callback survey. The differences between the population estimates of the respondents and non-respondents or the combined frequency rates represent the attitudinal differences between the groups, which also demonstrates whether volunteer bias is present in the initial survey results.

The unbiased estimates of the variances of the population means $\bar{y}_d$ were calculated as:

$$\widehat{\text{var}}(\bar{y}_d) = \frac{N-1}{N} \sum_{h=1}^L \left(\frac{n_h' - 1}{n' - 1} - \frac{n_h - 1}{N - 1} \right) \frac{w_h s_h^2}{n_h} + \frac{N - n'}{N(n' - 1)} \sum_{h=1}^L w_h (\bar{y}_h - \bar{y}_d)^2$$

where N is the total population (contact register); h is the stratum (1 = initial survey, 2 = callback survey), n_h , $w_h = n_h/n'$ and s_h^2 are the number of respondents, the previously specified weighting, and the estimated standard deviation of y in stratum h (Thompson, 2012, p. 193). The standard errors were calculated as the square root of provided variances:

$$s_d = \sqrt{\widehat{\text{var}}(\bar{y}_d)}$$

which were also used to calculate the 95% confidence intervals and their upper and lower boundaries:

$$\bar{y}_d \pm z s_d$$

where $z = 1.96$ is the 0.975th standard normal quantile resulting in two-tailed 95% confidence intervals.

4. Results

4.1. Forest owners

The unbiased estimates (combined frequency rates) showed that nearly 42% ($\pm 8.7\%$ standard error) of forest owners were interested in implementing GOFM in their forests, and approximately the same proportion ($40.0 \pm 8.9\%$) had implemented or were currently implementing GOFM (Table 2). More than 60% ($\pm 8.8\%$) of forest owners were aware of GOFM before taking part in the survey, and a high proportion of forest owners ($58.3 \pm 8.5\%$) considered GOFM information to be readily available, and less than 5% ($\pm 0.6\%$) of forest owners believed the opposite. Around 37% ($\pm 8.5\%$) of forest owners were entirely unaware of the availability of GOFM information.

Nearly 40% of forest owners were willing to receive more information about GOFM (Table 2). In addition, more than one third of forest owners ($35.1 \pm 8.1\%$) considered that his/her GOFM knowledge level influenced their motivation to implement the method. Only one fifth of forest owners ($21.9 \pm 7.3\%$) acknowledged the availability of forestry services that take care of game-oriented forest planning, consulting, management operations and other forest procedures. Less than 20% ($\pm 7.3\%$) of forest owners considered that such services are not available, and the majority ($59.8 \pm 8.9\%$) seemed to be unaware of the GOFM services. In addition, just over 5% ($\pm 0.6\%$) of forest owners admitted that they have currently been offered GOFM services, in the form of guidance and education.

As expected, attitude-related volunteer bias was present in the initial survey data, and the respondent group was clearly more invested in GOFM than the non-respondents. The respondents were more interested in GOFM (> 27 percentage point, pp), often thought that there was not sufficient information available (< 25 pp), were more willing to gain further information (> 19 pp), and more often believed that their GOFM knowledge level affected their subsequent motivation to implement GOFM (> 26 pp) compared to the non-respondents (Table 2). A small proportion (16%) of respondents also had experience of GOFM services, whereas the non-respondents had no GOFM experience at all. The smallest differences between respondents and non-respondents were observed among topics related to forest owners' awareness of GOFM, experiences of GOFM and opinions about the availability of services (Table 2). However, minor differences observed in regard to the results of the previous questions do not necessarily indicate that the respondents and non-respondents are equally interested in GOFM, especially as the greater interest in GOFM by the respondents seems obvious in regard to all other results.

4.2. Forest professionals

The unbiased estimates (combined frequency rates) showed that the concept of game management within forestry operations was familiar to most forest professionals, as $99\% (\pm 0.2\%)$ had previous knowledge of GOFM (Table 3). Also, a high proportion of professionals ($64.9 \pm 7.8\%$) believed that GOFM-related information was widely available, in contrast to the views of the forest owners (Tables 2 & 3). Nearly all professionals ($90.9 \pm 3.9\%$) also considered that more GOFM information should be offered to forest owners.

Almost $75\% (\pm 7.5\%)$ of professionals assumed that GOFM or its promotion was part of their normal duties or job description (Table 3). Approximately $90\% (\pm 5.3\%)$ of professionals had promoted GOFM in their work, and a higher proportion ($97.9 \pm 0.2\%$) was keen to promote GOFM. In addition, $80\% (\pm 6.4\%)$ of professionals considered their GOFM-related knowledge to be adequate in their job descriptions and everyday work. More than one third of professionals ($40.5 \pm 7.5\%$) were interested in receiving more GOFM information and further related education and consultation, and almost $60\% (\pm 7.8\%)$ of professionals had searched for information via the internet or had independently taken related courses. Less than one quarter of professionals ($23.6 \pm 6.3\%$) had received GOFM education from their employer.

In regard to attitude-related volunteer bias, the differences between respondents and non-respondents were minor in regard to most topics, but as expected, respondents tended to consider GOFM as part of their work to a greater extent than the non-respondents (> 16 pp), and respondents appeared to be more willing to take on more GOFM information (> 24 pp) than non-respondents. Also, a greater number of respondents had received GOFM education from their employer, in comparison to non-respondents (> 12 pp). The mostly minor differences between the unbiased estimates and the results of the initial survey in the forest professionals' group might indicate that the professionals feel obligated to respond to this type of survey ex officio.

5. Discussion

5.1. Adaptation of GOFM and its implementation

Based on our results, the number of forest owners interested in GOFM or in implementing GOFM seems modest. A rough comparison of our results with Rogers' adopter categorization (1995), would suggest that GOFM has been adopted by innovators, early adopters and even a group of early majority adopters among forest owners. However, to date, the greater proportion of forest owners, the late adopters (see Korhonen et al., 2013), have not adopted GOFM, which would suggest that recently provided GOFM information has not functioned as anticipated in encouraging forest owners to implement GOFM. This gives insights to our main research objectives, and is supported by Schultz (2011) and Heberlein (2012), who have observed that increased education does not necessarily lead to change in behavior (see also Fishbein and Ajzen, 2010), and core beliefs are often resistant to change in response to provided information (Sabatier and Weible, 2007).

High awareness of GOFM (more than 60%) does not reveal how much the forest owners actually know about GOFM, and low knowledge level might be one reason hindering the adoption of the method. Dayer et al. (2016) have noted that the poor participation of forest owners in incentive programs that aim to promote early successional forest habitats (ESH) is related to the fact that the forest owners may not understand the full value of creating these habitats, which are valuable to many wildlife species. In this regard, it is necessary to discuss the content of the provided GOFM information. For example, Gobster (2001), Laitila et al. (2009) and Dayer et al. (2016) have suggested that outreach activities should specifically emphasize the benefits and intentions of the promoted management practice at issue. It is possible that currently available information does not include elements that appeal to forest owners, which has led to forest owners being uninterested in GOFM, and may also explain why forest owners do not see their GOFM knowledge as affecting their motivation to implement GOFM. Our double sampling method provided us with results that can be generalized more reliably to the entire population of Finnish forest owners. Moreover, the strong volunteer bias in the results of the initial survey also suggests that the majority of forest owners are not as interested in GOFM as the forest owners that responded to the initial survey. The majority of forest owners, especially the later adopter groups, require convincing that the norms favor innovation, and that actual economic benefits can be expected from the adoption and application of an innovation (Rogers, 1995; Korhonen et al., 2013). In practice, GOFM has been considered cost-efficient and undemanding, although there is little scientific evidence to support this observation, even if research has begun to focus more on GOFM.

On the other hand, the low level of GOFM implementation might be because the information that is currently available appears insufficient for those forest owners who are more interested/willing to implement GOFM (respondents) (see e.g. Dayer et al., 2016). Yet, even if suitable information were provided, it would not necessarily result in the adoption of GOFM by the "passive" forest owners. Indeed, it has been suggested that for the efficient implementation of forest management related innovations, forest owners should be provided with tangible examples, demonstration sites and practical experiences from peer forest owners, in order to be able to learn, exchange information and observe the quantifiable benefits of these actions (Kittredge, 2004; Korhonen et al., 2013). However, these methods have not yet been used to the best of our knowledge. The unwillingness of forest owners to receive more GOFM information (as observed here) could be overcome with these strategies, where social benefits and positive experiences may enhance willingness to gain more knowledge about the topic or may stimulate emotions, which could subsequently affect attitudes and behavior in a positive manner and thus enhance the adoption of GOFM. Even so, it must be acknowledged that previously experienced negative associations might also hinder adoption, despite available information.

Table 3

Results from the game-oriented forest management (GOFM) questions posed to forest professionals. $\bar{y}_d$ is the unbiased estimate of the population mean (combined frequency), and s_d its standard error. CI, LB and CI, UB represent the lower and upper bounds of confidence interval (95%) of the population mean $\bar{y}_d$. On the right-hand side, $\bar{y}_1$ (respondent population mean) represents the proportion of responses in the initial survey and $\bar{y}_2$ (non-respondent population mean) the proportion of responses in the callback survey.

		$\bar{y}_d$, %	s_d , %	CI, LB	CI, UB	$\bar{y}_1$, %	$\bar{y}_2$, %
Are you aware of GOFM?	Yes	99.0	0.2	98.7	99.3	97.3	100.0
	No	1.0	0.2	0.7	1.3	2.7	0
Is GOFM related information available?	Yes	64.9	7.8	49.6	80.3	69.0	62.5
	No	16.8	6.3	4.4	29.2	13.4	18.8
	I don't know	18.3	6.3	5.9	30.7	17.6	18.8
Should forest owners be offered more GOFM information or support?	Yes	90.9	3.9	83.3	98.6	86.2	93.8
	No	9.1	3.9	1.4	16.7	13.8	6.3
Do you consider GOFM to be a part of your job or can it be linked to your work?	Yes	74.8	7.5	60.1	90.0	85.3	68.8
	No	23.9	7.5	9.2	38.7	11.3	31.3
	I don't know	1.3	0.2	0.9	1.6	3.4	0
Have you promoted GOFM in your work when reasonable?	Yes	88.0	5.3	77.6	98.5	88.9	87.5
	No	12.0	5.3	1.5	22.4	11.1	12.5
Would you like to promote GOFM in your work?	Yes	97.9	0.2	97.4	98.4	94.4	100.0
	No	2.1	0.2	1.6	2.6	5.6	0
Do you consider your GOFM-related knowledge adequate in your work?	Yes	80.0	6.4	67.5	92.4	77.8	81.3
	No	20.0	6.4	7.6	32.5	22.2	18.8
Would you like to have more GOFM information, education or guidance?	Yes	40.5	7.5	25.9	55.1	55.8	31.3
	No	59.5	7.5	44.9	74.2	44.2	68.8
Have you searched for GOFM information or education from alternative sources (e.g. internet, courses)?	Yes	59.4	7.8	44.1	74.8	54.2	62.5
	No	40.6	7.8	25.2	55.9	45.8	37.5
Have you received GOFM-related education from your employer?	Yes	23.6	6.3	11.2	35.9	31.7	18.8
	No	76.4	6.3	64.1	88.8	68.3	81.3

Uncertainties related to GOFM services and the lack of GOFM guidance, which seem to highlight a missing interaction between forest owners and professionals, are likely obstacles for GOFM implementation. However, the poor service selection experienced by forest owners does not seem to be caused from a lack of knowledge, competence or promotion on the part of the forest professionals. Moreover, the effect of attitude-related volunteer bias on the professionals' results was not as pronounced as with forest owners (although sufficiently to reflect a need for bias correction), which suggests that no obvious attitude issues exist in relation to GOFM among professionals. Here, professionals can be recognized as change agents, who facilitate innovations (GOFM in this case) to forest owners as clients (see [Muth and Hendee, 1980](#); [Hubbard and Sandmann, 2007](#); [Korhonen et al., 2013](#)). Change agents help the clients to become aware of the need to change their behavior, and influence them to take corrective actions ([Rogers, 1995](#)), although this does not seem to have succeeded in the case of GOFM. [Korhonen et al. \(2013\)](#) found similar results, stating that the dissemination of information to “passive” forest owners might not succeed if it is only channeled through professionals; it also requires active learning and the experience of forest owners, which could be enhanced with the help of professionals.

Overall, the strong volunteer bias present in the results of the forest owners' initial survey and the more subtle bias in the results of the professionals, suggest that generalization of these results would have led to misleading conclusions of the state of GOFM implementation. For example, a naïve estimate for the proportion of forest owners interested in GOFM would have been 60.6% with a 95% confidence interval of 55.8–65.4%. Taking into account the discrepancy between respondents and non-respondents, volunteer bias was corrected and a more reliable estimate for the proportion of GOFM-interested forest owners was found to be 41.9% ($\pm 8.7\%$).

5.2. Institutional adaptation and professional norms

The imbalance between awareness of GOFM and the measures put into practice among forest owners is likely linked to the uncertainties they have experienced with the availability of GOFM information and services. Innovations in the forest sector, such as GOFM, might be hindered by established organizations, which try to maintain their positions in the production-driven markets ([Mattila et al., 2013](#); [Mattila and Roos, 2014](#)), and this could have affected the visibility of current GOFM services. Decisions regarding habitat conservation are generally made as components of traditional forest planning and execution ([Primmer and Wolf, 2009](#)). Indeed, GOFM might have become assimilated within traditional forestry approaches, with [Primmer \(2011\)](#) suggesting that biodiversity conservation of non-industrial private forests is built in to common forest management, and that mainstream forestry has subsumed it. This would suggest that GOFM-related options are in fact available, as the results of the professionals survey would suggest, but that these are currently tightly integrated into mainstream forestry and no specialized GOFM services are provided (see e.g. [Primmer, 2011](#)). Therefore, the forest owners that are aware of GOFM, and that are potentially interested in GOFM might have failed to find a GOFM-oriented service or advice provider (see also [Mattila and Roos, 2014](#)), which might have impeded the generalization of GOFM.

In regard to the adoption of GOFM at the professionals level, the results that show an absence of GOFM education seem meaningful, although professionals are already qualified in GOFM. For example, organizational culture, support of management, and changing circumstances and priorities at a workplace may affect the appliance of an individual's knowledge and skills obtained from vocational courses ([Draper et al., 2014](#)), and the production-oriented field has likely suppressed the utilization of GOFM skills of the professionals. [Primmer](#)

(2011) has suggested that the forest sector should assist forest actors in modifying their actions, utilizing new approaches or sharing their practices with each other in order to promote biodiversity conservation in the private forests that are managed commercially. In this way, the GOFM skills of the professionals would also be better utilized. This type of education might enable the promotion of GOFM inside working organizations and emphasize the role of efficient change agents not only in the adoption of GOFM among forest stakeholders, but also at the level of institutional adaptation. Changes in the freedom to operate among innovation goals and the addition of education that enables professionals to interactively share information with each other could result in an increase in the number of specialized GOFM services provided.

5.3. Suitability of current forest policy approach

In Finland, previous examples of voluntary-based forest management schemes have shown promise and have led to fruitful outcomes (e.g. Paloniemi and Varho, 2009), and adoption of GOFM might still be ongoing. An important feature of GOFM has been that forest owners do not need to register in any particular scheme, as GOFM is a series of management options available for forest owners to use freely. This weakens the possibilities to assess the application of GOFM (some might apply GOFM-like methods and not consider it as GOFM) and ways to evaluate the effectiveness of this voluntary-based approach. Currently, the rather high awareness of GOFM would suggest that available information and the new possibilities to implement GOFM (introduced after 2014) have also reached adopter groups other than innovators and early adopters, but has not yet led to wider acceptance. However, it might be too early to make assumptions of the suitability of the voluntary-based approach in GOFM promotion, especially as it seems that alternative knowledge dissemination methods have not been widely tested to provide GOFM information to forest owners. Either way, it is reasonable to discuss the usability of other approaches in the generalization of GOFM.

As the most recent changes in forest legislation have endeavored to increase the freedom of choice and responsibilities of forest owners in forest management (Ojala and Mäkelä, 2013), it is unlikely that GOFM will become obligatory. Moreover, a shift to policies that promote this type of forest ecosystem service will require a decrease in the economic importance of the main product from forests (roundwood) (Ingold and Zimmermann, 2011). One possible approach would be to include GOFM more explicitly in forest certification criteria, which would ensure that forests under certificate are managed and harvested in accordance with the criteria, and that forest owners would be formally committed to GOFM, thereby initiating a change in the operational culture. Despite certification systems are voluntary, they might help in the generalization of GOFM, as most forest owners have committed to the certification of their forests.

Other possible approaches could be related to economic mechanisms, and of course, GOFM-related incentives or compensations might positively affect the motivation of forest owners to implement GOFM. However, as there is no prepared incentive system for GOFM, this approach would require more research, as many of the previous incentive-based approaches are considered ineffective (e.g. Bieling, 2004; Kilgore et al., 2007; Dayer et al., 2014). In regard to forest management decision-making, forest owners appreciate one-to-one guidance over incentives (Kilgore et al., 2007), which puts more pressure on the advice providers, and this may also be the case for the generalization of GOFM. As a consequence, (and due to the informal nature of GOFM), it might be reasonable to consider a more practical approach to GOFM knowledge dissemination, and provide forest owners with more tangible GOFM experiences and interactive peer learning before modifying policy instruments that are based on either obligatory or economic approaches.

5.4. Conclusions and future implications

There seems to be a moderate level of GOFM implementation by forest owners, and late adopters have not yet accepted GOFM. Moderate implication levels might be related to the fact that forest owners do not have sufficient knowledge of GOFM, or later adopters' higher requirement for persuasion for adopting. Wider implementation of GOFM might require the provision of information that emphasizes the benefits of GOFM, and as game species have been considered as good justification for altering forest management approaches, studying the effects of GOFM on game populations might possibly encourage forest owners to implement GOFM. However, peer pressure may be necessary, as affirmation of the suitability of the method must come from peer forest owners. It might be reasonable to test these type of knowledge dissemination methods before assessing the suitability of the current voluntary approach for GOFM implementation.

The high engagement and high-grade knowledge and competence of professionals in GOFM seems to reflect that professionals are already adjusting to GOFM implementation in the operational field. Nevertheless, the mismatch between awareness of GOFM and its implementation among forest owners, and the forest owners' unawareness of the availability of GOFM services, seems to indicate that organizations have not yet succeeded in facilitating a ground where forest owners are able to adopt and implement GOFM. Despite many improvements and much effort (e.g. possibilities to reserve game thickets in private forests), GOFM seems to be merged into traditional forest management and services, which reflects a developing institutional adaptation to GOFM. However, as our study evaluates the perspectives of individual forest professionals and their working organizations, we cannot make detailed assumptions about institutional adaptation to GOFM from a wider standpoint (forest sector and operational networks). At the level of professionals, improved education possibilities could strengthen their role as change agents and help to disseminate knowledge inside institutions, help to improve the market services that are already available, or assist in generating GOFM services that reach forest owners. Education should acknowledge the requirement of peer networking in the generalization of GOFM among forest owners, as professionals might need additional tools to create other approaches to disseminate GOFM information, as well as maximize the interaction in information exchange between forest owners.

Overall, the differences between the unbiased estimates and the initial survey were sufficient to indicate that the double sampling method employed in this study was successful in providing more reliable results than those obtained with simple random sampling.

Declaration of Competing Interest

None.

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